

need to be aware of the correct spelling for R function names and objects:

```
> Shingle()
Error in Shingle() : could not find function "Shingle"
```

Unmatched parenthesis

An unmatched or extra parenthesis in an R expression is reported as an error. You should use your editor settings to identify such syntax errors before running the code:

```
> shingle(mtcars$cy1))
Error: unexpected ')' in "shingle(mtcars$cy1))"
```

Subscript out of bounds

Consider the following multi-dimensional array:

```
> a <- array(c(1, 2, 3, 4, 5, 6, 7, 8, 9), dim=c(3, 3))
> a
      [,1] [,2] [,3]
[1,]    1    4    7
[2,]    2    5    8
[3,]    3    6    9

> a[1,3]
[1] 7
```

An array access beyond the memory bounds throws the 'subscript out of bounds' error:

```
> a[1,4]
Error in a[1, 4] : subscript out of bounds
```

Invalid subscript type list

The *mtcars* data has multiple columns. You can store a select list of column names to a variable and reference them using the *rowSums()* function. But to access the individual list elements, the *unlist()* function needs to be used, as demonstrated below:

```
> x <- list("mpg", "cyl", "disp")

> rowSums(mtcars[,c(x)])
Error in .subset(x, j) : invalid subscript type 'list'

> rowSums(mtcars[unlist(x)])
      Mazda RX4      Mazda RX4 Wag      Datsun 710
Hornet 4 Drive
      187.0          187.0          134.8
285.4
Hornet Sportabout      Valiant      Duster 360
```

```
Merc 240D
      386.7          249.1          382.3
175.1
      Merc 230          Merc 280          Merc 280C
Merc 450SE
      167.6          192.8          191.4
300.2
      Merc 450SL      Merc 450SLC  Cadillac Fleetwood
Lincoln Continental
      301.1          299.0          490.4
478.4
      Chrysler Imperial      Fiat 128      Honda Civic
Toyota Corolla
      462.7          115.1          110.1
109.0
      Toyota Corona      Dodge Challenger      AMC Javelin
Camaro Z28
      145.6          341.5          327.2
371.3
      Pontiac Firebird      Fiat X1-9      Porsche 914-2
Lotus Europa
      427.2          110.3          150.3
129.5
      Ford Pantera L      Ferrari Dino      Maserati Bora
Volvo 142E
      374.8      170.7      324.0      146.4
```

Undefined columns selected

A typo in the column names in a data set can lead to an 'undefined columns' error, as shown below:

```
> mtcars[,c("display")]
Error in `[.data.frame' (mtcars, , c("display"))` :
  undefined columns selected

> mtcars[,c("disp")]
[1] 160.0 160.0 108.0 258.0 360.0 225.0 360.0 146.7 140.8
167.6 167.6 275.8
[13] 275.8 275.8 472.0 460.0 440.0 78.7 75.7 71.1 120.1
318.0 304.0 350.0
[25] 400.0 79.0 120.3 95.1 351.0 145.0 301.0 121.0
```

Non-numeric argument to a binary operator

The *sum()* function is used to compute the sum of numeric values. If it is used with a string object, R reports a 'non-numeric argument' error.

```
> a <- "Hello World"

> a
[1] "Hello World"
```